# ROADRESQ
# Payment & Settlement Specification

Customer Payment → Platform Ledger → Provider Payout → Refund → Reconciliation

**Version:** 1.0 • **Status:** Production financial operating baseline

AUTHORITATIVE MONEY RECORDS • IDEMPOTENCY • RECONCILIATION • AUDITABILITY

**Scope:** Customer payments • platform fees • provider earnings • payouts • refunds • disputes • settlement

# Document Control

| Field | Value |
|---|---|
| Document | RoadResQ Payment & Settlement Specification |
| Version | 1.0 |
| Status | Production operating baseline |
| Owners | Finance + Engineering + Product |
| Scope | Customer collection, internal ledgering, provider settlement, refunds, payout operations and reconciliation |
| Related documents | API Specification • Database/Data Dictionary • Security & Threat Model • Customer Support & Dispute SOP • Production Runbook • Backup/DR |
| Core principle | Never infer money state from UI state; use authoritative transaction records and reconciliation. |

# 1. Purpose

This specification defines the financial lifecycle for RoadResQ from customer payment through platform accounting, provider earnings, refunds and payout settlement. It establishes authoritative records, state transitions, controls and reconciliation processes required to operate a production marketplace safely.

• Every monetary movement must have an auditable source and destination.

• Customer payment status and provider earnings must derive from authoritative records.

• Duplicate payment, refund and payout actions must be prevented with idempotency and state controls.

• Provider payout must not occur before the booking reaches the configured settlement-eligible state.

• Refunds must be linked to the original customer payment.

• Financial discrepancies must enter reconciliation rather than being silently overwritten.

• Payment credentials and sensitive financial data must remain within approved security boundaries.

# 2. Financial Actors

| Actor | Responsibility |
|---|---|
| Customer | Pays for approved service/invoice using supported methods. |
| RoadResQ | Records payment/ledger state, applies configured fees and coordinates refunds/settlement. |
| Provider | Earns eligible service amount after applicable fees/adjustments. |
| Payment Gateway | Authorizes/captures payments and processes refunds. |
| Bank / Payout Rail | Transfers provider settlement funds. |
| Finance Operations | Reconciles transactions, manages settlement and financial exceptions. |
| Support | Handles customer payment/refund cases within authorized limits. |

# 3. Core Money Flow

**Customer** → payment gateway → **RoadResQ payment record** → **invoice/ledger** → platform/provider allocation → **provider payable balance** → **payout**.

Refunds reverse or adjust eligible customer-side financial entries and must always reference an original source transaction.

# 4. Financial Objects

| Object | Purpose |
|---|---|
| Invoice | Authoritative customer bill for approved/completed service. |
| Invoice item | Individual charge component. |
| Payment | Customer payment attempt/transaction linked to invoice/booking. |
| Refund | Full or partial reversal of an eligible payment. |
| Ledger entry | Append-only/versioned financial movement or allocation. |
| Provider earning | Amount allocated as payable to provider. |
| Payout | Settlement transfer to provider. |
| Payout item | Individual earning included in a payout. |
| Adjustment | Authorized fee, credit, reversal or correction. |
| Reconciliation record | Comparison between internal and external financial records. |

# 5. Monetary Amount Model

Authoritative money values should use integer minor units, such as paise, with an explicit ISO currency code. Floating-point arithmetic must not be used for financial calculations.

• Use deterministic server-side calculations.

• Use one documented rounding policy.

• Represent reversals/credits through typed ledger entries.

• Store currency explicitly on every financial object.

• Convert minor units to localized currency only for presentation.

# 6. Payment Lifecycle

| State | Meaning |
|---|---|
| CREATED | Payment intent/order created. |
| PENDING | Payment started or gateway confirmation pending. |
| AUTHORIZED | Payment method authorized where gateway supports separate authorization. |
| CAPTURED | Funds successfully captured. |
| FAILED | Payment attempt failed. |
| CANCELLED | Payment cancelled before capture where supported. |
| PARTIALLY_REFUNDED | Part of captured amount refunded. |
| REFUNDED | Full eligible amount refunded. |
| DISPUTED | Chargeback/external or internal payment dispute active. |
| RECONCILIATION_HOLD | Financial state requires reconciliation before further settlement. |

# 7. Customer Payment SOP

• Create an invoice/payment intent from authoritative booking and pricing data.

• Generate a unique payment/order identifier.

• Use idempotency for payment creation where supported.

• Use the approved gateway checkout/tokenized flow.

• Treat client-side success screens as non-authoritative.

• Confirm payment through verified gateway response/webhook and server-side checks.

• Store gateway transaction/reference ID.

• Set CAPTURED only after authoritative confirmation.

• Link captured payment to invoice and booking.

• Create provider earning allocation only according to settlement rules.

# 8. Payment Security Controls

• Do not store raw card numbers, CVV or restricted payment credentials unless explicitly required and permitted by the chosen architecture.

• Prefer gateway-hosted/tokenized payment flows.

• Verify webhook signatures.

• Use TLS/HTTPS for payment communication.

• Use idempotency keys for supported operations.

• Restrict financial actions with RBAC and audit logging.

• Mask sensitive payment identifiers in logs.

# 9. Invoice & Approval Rules

• Invoice amount must originate from approved estimate/job data and configured business rules.

• Additional work must require customer approval where applicable.

• Post-payment changes must use versioned adjustments rather than silently rewriting history.

• Final invoice and collected amount must agree.

• Taxes/fees should be separately represented where required.

• Provider earnings derive from the final authoritative financial record.

# 10. Platform Ledger

The internal ledger is the source of truth for allocations and settlement calculations. The external gateway remains authoritative for gateway-side transaction state; both systems must be reconciled.

• Ledger entries are append-only or versioned.

• Each entry references a booking/invoice/payment where applicable.

• Record amount, currency, type, timestamp, actor/system and correlation ID.

• Corrections use adjustment/reversal entries.

• Provider payable balance is reproducible from eligible earnings minus holds, adjustments and payouts.

# 11. Provider Earning Lifecycle

| State | Meaning |
|---|---|
| PENDING_EARNING | Potential provider amount exists but is not payout eligible. |
| ELIGIBLE | Settlement conditions have passed. |
| ON_HOLD | Temporarily blocked by dispute, refund, risk or operational condition. |
| SCHEDULED | Included in a payout batch. |
| PAYOUT_PROCESSING | Payout submitted. |
| PAID | Payout confirmed. |

| State | Meaning |
| --- | --- |
| PAYOUT_FAILED | Transfer failed and requires review/retry. |
| ADJUSTED | Prior earning changed by authorized financial adjustment. |

# 12. Provider Payout Eligibility

• Booking is in the configured completed/settlement-eligible state.

• Required invoice/payment conditions are satisfied.

• No blocking dispute/refund/risk hold exists unless policy allows partial settlement.

• Provider payout information is valid.

• Provider remains eligible under provider operations rules.

• Fees, taxes and authorized adjustments are calculated before payout.

• Settlement batch has a unique identifier and auditable contents.

# 13. Payout Batch Process

• Identify provider earnings eligible for settlement.

• Apply holds and adjustments.

• Generate payout items and calculate net provider payout.

• Validate batch totals.

• Lock the batch against accidental duplication.

• Submit through the approved payout mechanism.

• Record external payout reference.

• Process authoritative success/failure response.

• Mark payout items PAID only after confirmation.

• Reconcile batch totals against external settlement records.

# 14. Payout Failure Handling

| Failure | Action |
| --- | --- |
| Invalid payout details | Hold payout and request provider correction. |
| Bank rejection | Record external reason; retry only under defined rules. |
| Timeout / unknown state | Check external status before retrying. |
| Duplicate risk | Use batch lock/idempotency and verify external reference. |
| Provider suspension | Hold payout according to policy and applicable obligations. |
| System error | Preserve payable balance and create incident if systemic. |
| Partial batch failure | Reconcile each payout item independently. |

# 15. Refund Lifecycle

| State | Meaning |
| --- | --- |
| REQUESTED | Refund request created. |
| APPROVED | Refund authorized. |
| PROCESSING | Refund submitted. |

| State | Meaning |
|---|---|
| SUCCEEDED | Gateway confirmed refund. |
| FAILED | Refund attempt failed. |
| PENDING_RECONCILIATION | Internal/external confirmation differs or is delayed. |
| REVERSED | Refund outcome changed according to external financial state. |

# 16. Refund Rules

• Every refund references the original payment.

• Full refund cannot exceed eligible captured amount.

• Cumulative partial refunds cannot exceed captured amount.

• Refund authorization should be separate from execution where practical.

• Use idempotency to prevent duplicate refund requests.

• Record refund reason, amount, currency, actor and timestamp.

• Do not issue another refund merely because gateway confirmation is delayed.

• Adjust provider earnings according to the configured settlement policy.

# 17. Refund Impact on Provider Settlement

| Situation | Provider financial treatment |
|---|---|
| Refund before provider payout | Reduce/remove eligible provider earning before payout. |
| Refund after provider payout | Create authorized provider balance adjustment/hold according to policy. |
| Partial refund | Apply configured allocation adjustment. |
| Platform-funded goodwill | Use approved platform adjustment where applicable. |
| Provider-responsible dispute | Apply provider-side adjustment only after evidence review/authorization. |
| Chargeback/payment reversal | Hold affected funds and reconcile before further payout. |

# 18. Chargebacks & External Disputes

• Record external dispute/chargeback reference.

• Link it to payment, invoice and booking.

• Place affected amounts on hold where required.

• Preserve invoice, approval, completion, communication and transaction evidence.

• Submit evidence through the approved dispute process where applicable.

• Do not create a duplicate refund while the external dispute is unresolved.

• Update internal state when the external outcome is confirmed.

# 19. Reconciliation Model

Reconciliation compares RoadResQ records with external payment and payout records to identify missing, duplicate, delayed or mismatched transactions.

| Layer | Compare |
|---|---|
| Payment | Internal payment vs gateway status, amount, currency and reference. |
| Refund | Internal refund vs gateway refund/status/amount/reference. |

| Layer | Compare |
| --- | --- |
| Payout | Payout batch/items vs payout rail records. |
| Ledger | Ledger totals vs payment/refund/payout source records. |
| Booking | Booking completion/invoice/payment vs settlement eligibility. |
| Settlement | Expected provider payable vs actual settled amount. |

## 20. Reconciliation Statuses

| Status | Meaning |
| --- | --- |
| MATCHED | Records agree. |
| PENDING | External confirmation expected. |
| AMOUNT_MISMATCH | Amounts differ. |
| STATUS_MISMATCH | States differ. |
| MISSING_EXTERNAL | Internal record has no external match. |
| MISSING_INTERNAL | External record has no internal match. |
| DUPLICATE | Multiple records represent the same transaction. |
| MANUAL_REVIEW | Finance investigation required. |

## 21. Daily Reconciliation SOP

• Retrieve authoritative gateway/payout records for the reconciliation window.

• Match using stable internal and external references.

• Compare status, amount, currency and relevant timestamps.

• Identify unmatched, duplicate and mismatched records.

• Resolve deterministic safe cases.

• Create manual review cases for ambiguous discrepancies.

• Confirm refunds and payout batches.

• Record totals and unresolved exceptions.

• Escalate high-value or systemic discrepancies.

## 22. Reconciliation Tolerance Rules

• Known asynchronous status delays may remain PENDING temporarily.

• Amount mismatches must not be silently tolerated.

• Currency mismatches are blocking.

• Duplicate transactions require investigation.

• Rounding differences are acceptable only under a documented currency/rounding rule.

• High-value mismatches require Finance review.

• Repeated mismatch patterns require Engineering/Product investigation.

## 23. Idempotency & Duplicate Protection

| Operation | Control |
| --- | --- |
| Payment creation | Unique internal ID + gateway idempotency key where supported. |

| Operation | Control |
| --- | --- |
| Capture | Idempotent request + authoritative status check. |
| Refund | Unique refund ID + idempotency key. |
| Payout batch | Unique batch ID + immutable payout-item membership. |
| Webhook | Store provider event ID; ignore already-processed events. |
| Ledger posting | Unique source-event/reference constraints. |
| Manual adjustment | Unique reference + elevated authorization. |

# 24. Webhook Processing

• Verify webhook authenticity/signature.

• Persist event before irreversible processing where architecture permits.

• Use provider event ID for idempotent processing.

• Do not assume webhook ordering.

• Handle delayed/out-of-order events through state reconciliation.

• Acknowledge valid events appropriately.

• Route malformed/suspicious events to monitoring/security.

• Restrict raw webhook payload access and redact sensitive data in logs.

# 25. Financial Holds

| Hold reason | Examples |
| --- | --- |
| Refund hold | Customer refund requested/processing. |
| Dispute hold | Active service/payment dispute. |
| Chargeback hold | External payment dispute. |
| Risk hold | Fraud/safety investigation. |
| Provider verification hold | Provider no longer meets required eligibility. |
| Payout failure hold | Invalid payout information or failed transfer. |
| Reconciliation hold | Internal/external state mismatch. |

# 26. Adjustments & Corrections

• Never edit historical financial records to make totals appear correct.

• Create typed adjustment/reversal records linked to the original transaction.

• Require appropriate authorization.

• Record reason, amount, currency, actor and evidence.

• Re-run affected settlement/reconciliation calculations.

• Large or unusual adjustments require elevated review.

# 27. Taxes, Fees & Platform Revenue

• Represent customer charges, taxes, platform fees and provider earnings as separate components where applicable.

• Use configurable fee rules rather than hard-coded assumptions.

• Record the rule/version used for material calculations.

• Pricing/fee changes must not rewrite historical transactions.

• Finance must validate tax/fee configuration against applicable requirements before launch.

# 28. Provider Statement

| Field | Description |
|---|---|
| Period | Statement start/end date. |
| Booking | Related service/booking reference. |
| Gross earning | Provider allocation before deductions. |
| Fees/adjustments | Platform fees, reversals and authorized adjustments. |
| Hold | Temporarily unavailable amount. |
| Net eligible | Amount available for settlement. |
| Payout | Associated payout reference/status. |
| Balance | Remaining provider payable balance. |

# 29. Customer Payment Receipt

• Booking/invoice reference.

• Paid amount and currency.

• Non-sensitive payment method description.

• Transaction/reference ID where appropriate.

• Payment status.

• Invoice breakdown.

• Refund amount/status/reference when applicable.

# 30. Financial Permissions

| Role | Typical access |
|---|---|
| Customer | View own invoices, payments and refunds. |
| Provider | View own earnings, payouts and statements. |
| Support | View limited payment/refund data; execute only authorized low-risk actions. |
| Finance | Reconciliation, settlement, refund review and authorized adjustments. |
| Admin | Operational oversight; elevated financial actions only when explicitly permitted. |
| Engineering | Technical troubleshooting without unnecessary financial mutation rights. |

# 31. Security & Privacy

• Apply least privilege to payment and payout operations.

• Separate viewing and mutation permissions where practical.

• Encrypt sensitive data in transit and at rest.

• Protect payout details and customer payment information.

• Audit refunds, adjustments, payouts and privileged access.

• Never expose payment secrets in support/admin screens.

• Follow the RoadResQ Security Architecture & Threat Model.

## 32. Settlement Calendar

Payout cadence must be configurable and approved before launch. Production configuration should define payout frequency, cut-off, holidays, minimum payout thresholds, holds and retry rules.

| Configuration | Definition |
|---|---|
| Frequency | Daily / weekly / configurable. |
| Cut-off | Configured local business time. |
| Minimum payout | Configured threshold if applicable. |
| Reserve/hold | Configured risk/dispute hold. |
| Retry | Defined retry behavior. |
| Holiday handling | Defined business-day/rail behavior. |
| Notifications | Payout initiated/success/failed notifications. |

## 33. Financial Incident Handling

• Duplicate charge: stop further collection, reconcile and refund verified duplicate amount.

• Duplicate refund: stop additional refund actions and escalate.

• Incorrect payout: calculate difference and create authorized adjustment.

• Gateway outage: preserve pending state and avoid blind retries.

• Webhook delay: reconcile external state before financial mutation.

• Ledger integrity concern: restrict financial mutation and activate incident response.

• Large unexplained discrepancy: escalate to Finance + Engineering/Security as appropriate.

## 34. Monitoring & Financial Alerts

| Alert | Trigger |
|---|---|
| Payment failure spike | Failure rate exceeds configured baseline. |
| Refund failure spike | Refund failures exceed threshold. |
| Webhook delay | Expected event remains unconfirmed beyond SLA. |
| Payout failure spike | Provider payout failures exceed baseline. |
| Reconciliation mismatch | Mismatch value exceeds threshold. |
| Duplicate detection | Potential duplicate payment/refund/payout. |
| Ledger imbalance | Financial totals cannot be reproduced. |
| High-value anomaly | Transaction exceeds configured review threshold. |

## 35. Financial Metrics

• Payment success and pending rates

• Refund success rate and processing time

• GMV / gross booking value

• Platform fee/revenue volume

• Provider payable balance

• Payout success/failure rate

• Settlement time

- Chargeback/dispute rate
- Reconciliation match rate
- Unresolved reconciliation value
- Manual adjustment volume

## 36. Critical Test Scenarios

- Successful customer payment.
- Payment failure.
- Successful payment with delayed webhook.
- Duplicate webhook.
- Out-of-order webhook.
- Customer retry after timeout without duplicate charge.
- Full refund and partial refund.
- Duplicate refund request.
- Pending/failed refund.
- Successful and failed provider payout.
- Unknown payout state after timeout.
- Payout retry without duplication.
- Provider suspension before payout.
- Chargeback after provider payout.
- Internal/external amount mismatch.
- Missing internal/external transaction.
- Manual adjustment with audit record.

## 37. Production Reconciliation Checklist

- ■ Gateway transaction feed available
- ■ Payout records available
- ■ Payment/reference matching completed
- ■ Refund matching completed
- ■ Payout batch matching completed
- ■ Duplicate detection completed
- ■ Amount/currency mismatch report reviewed
- ■ Missing internal/external transactions reviewed
- ■ High-value exceptions escalated
- ■ Ledger totals verified
- ■ Unresolved reconciliation value recorded
- ■ Finance sign-off completed

## 38. Production Definition of Done

- Customer payment lifecycle uses authoritative gateway confirmation.
- Payments are linked to invoices and bookings.

- Money calculations use minor units and explicit currency.
- Ledger records are immutable/versioned and auditable.
- Provider earnings and payout states are implemented.
- Payout batches are idempotent and auditable.
- Full and partial refunds link to source payments.
- Payment/refund/payout webhooks are verified and idempotent.
- Reconciliation identifies mismatches, duplicates and missing records.
- Financial holds and adjustments are supported.
- RBAC protects financial operations.
- Customer/provider financial statements are understandable.
- Monitoring covers critical financial failures.
- QA covers payment, refund, payout and reconciliation edge cases.
- Finance has a documented reconciliation process before production launch.

# 39. Final Financial Standard

RoadResQ should treat money movement as a stateful, auditable system rather than a collection of payment buttons. The authoritative flow is: **customer payment** → **verified transaction** → **internal ledger** → **provider earning** → **settlement eligibility** → **payout** → **reconciliation**. Refunds, disputes and corrections must always connect back to their original financial records. No UI state, retry or manual action should be able to create an untraceable or duplicate movement of money.